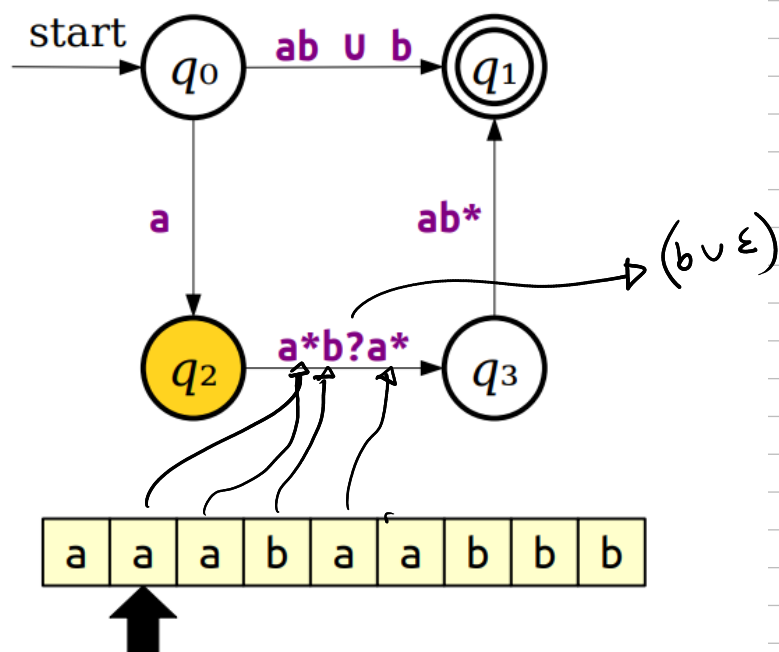


DFA to RE



$$a^* = \{\epsilon, a, aa, aaa, \dots\}$$

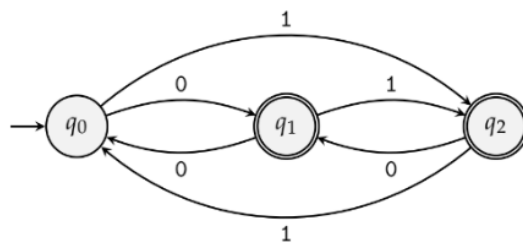
Notes:

State elimination

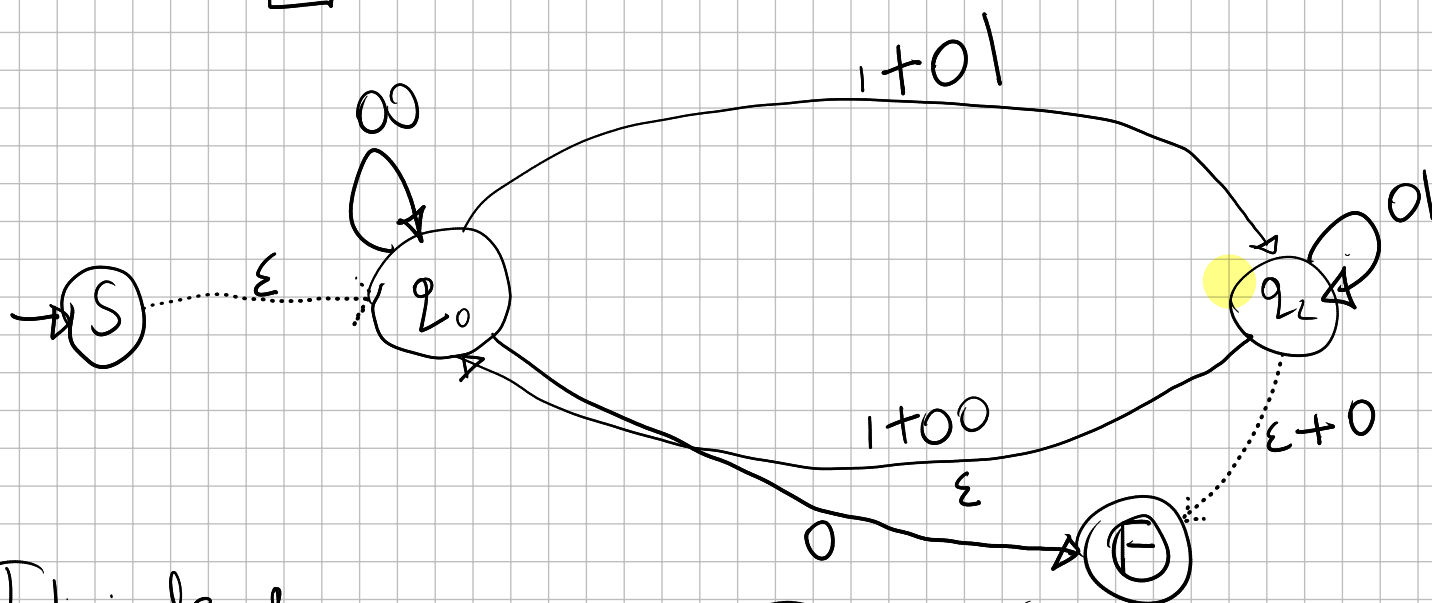
- (i) You will be give sequence of states you need to eliminate.
- (ii) If start state has incoming edge/transition, Create a new start state.
- (iii) Always Create a new final state that extends from previous final states.

Problem 4 (CO3): Converting Finite Automata to Regular Expressions (10 points)

Convert the following DFA into an equivalent regular expression using the state elimination method. First eliminate q_1 , then , and finally q_0 . You must show work.

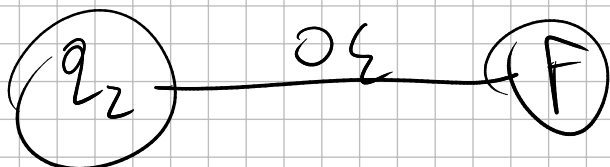
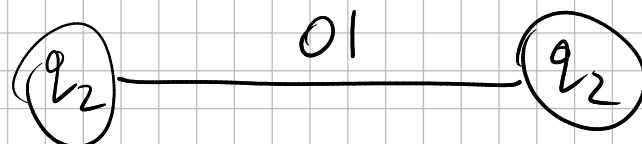
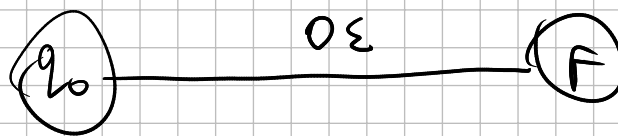
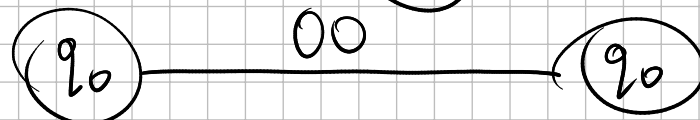
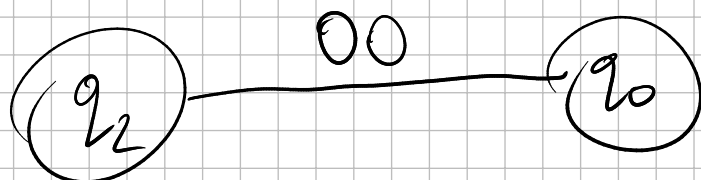


Eliminate q_1, q_2, q_0



Eliminate q_1

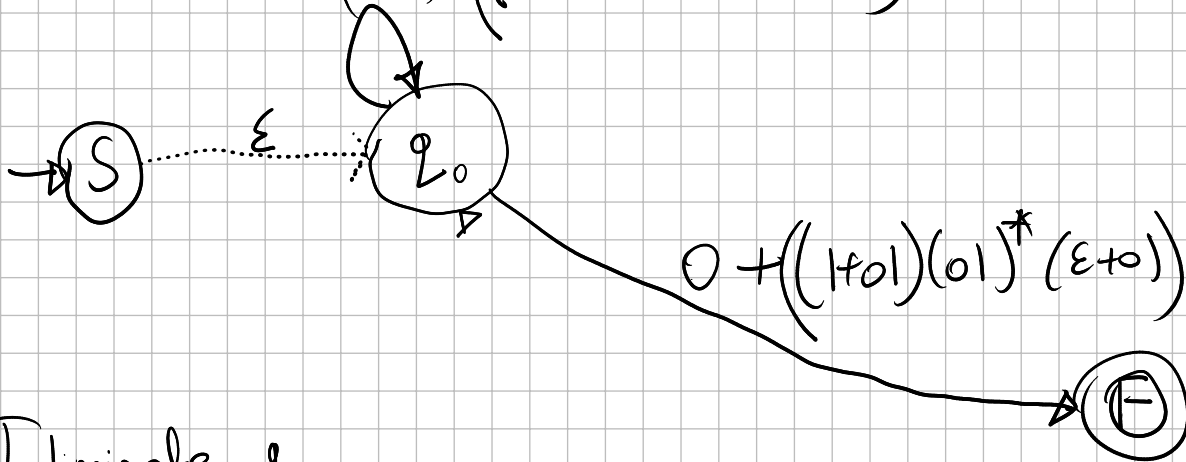
In	out
q_0	q_0
q_2	q_2
q_0	F
q_2	F



Eliminate $q_1, \boxed{q_2}, q_0$

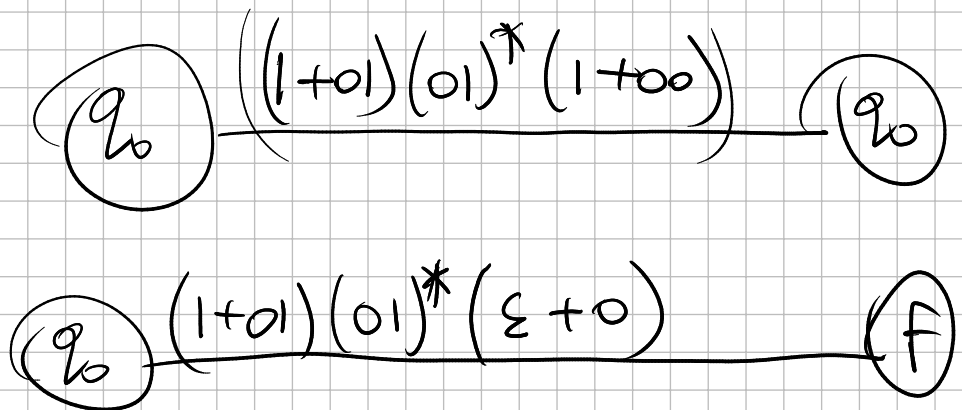
$$(01)^* = \{\varepsilon, \underline{01}, \underline{0101}, \dots\}$$

$$(00) + ((1+01)(01)^*(1+00))$$

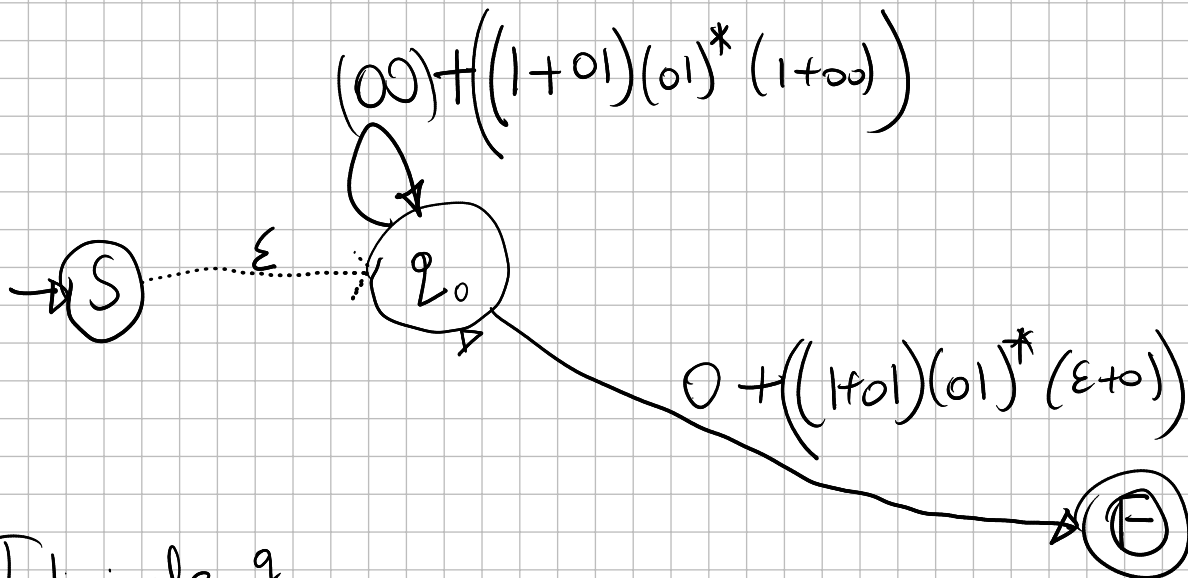


Eliminate q_2

In	out
q_0	q_0
	F

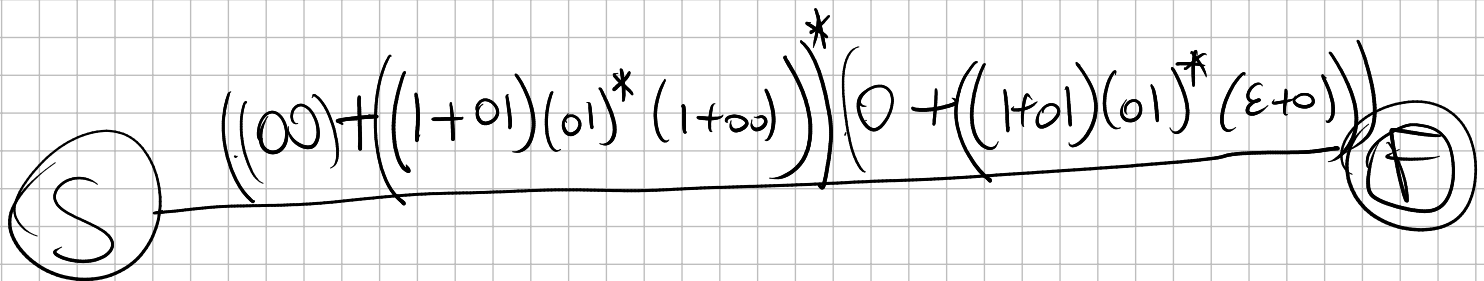


Eliminate $q_1, q_2, \boxed{q_0}$



Eliminate q_0

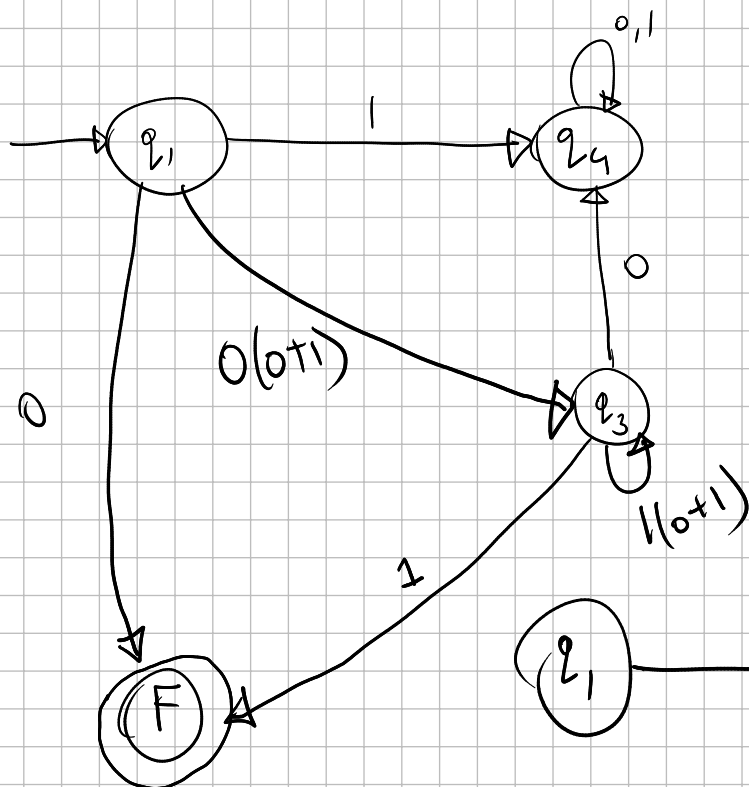
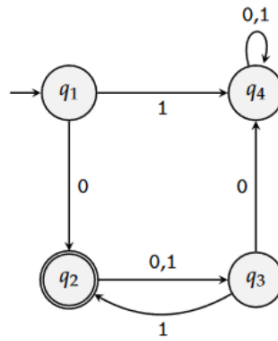
In	out
S	F



$$((0) + ((1+01)(01)^*(1+00)))^*(0 + ((1+01)(01)^*(1+00)))$$

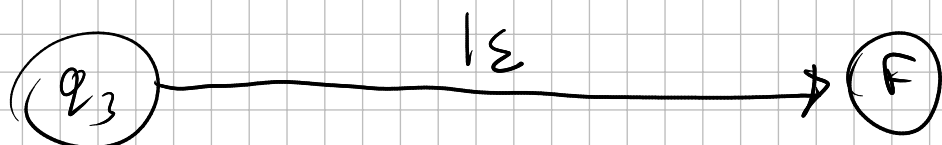
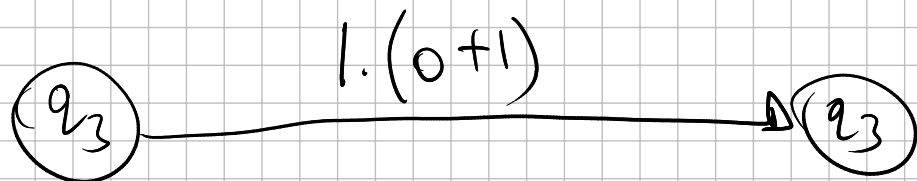
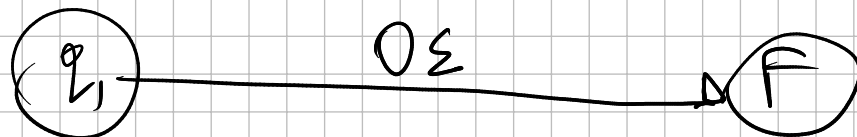
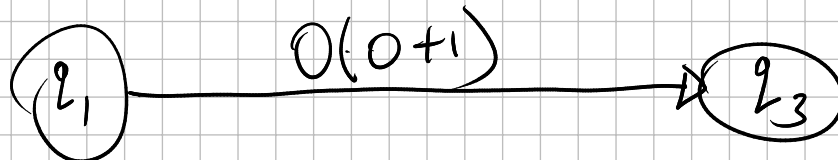
Problem 4 (CO2): Converting Finite Automata to Regular Expressions (10 points)

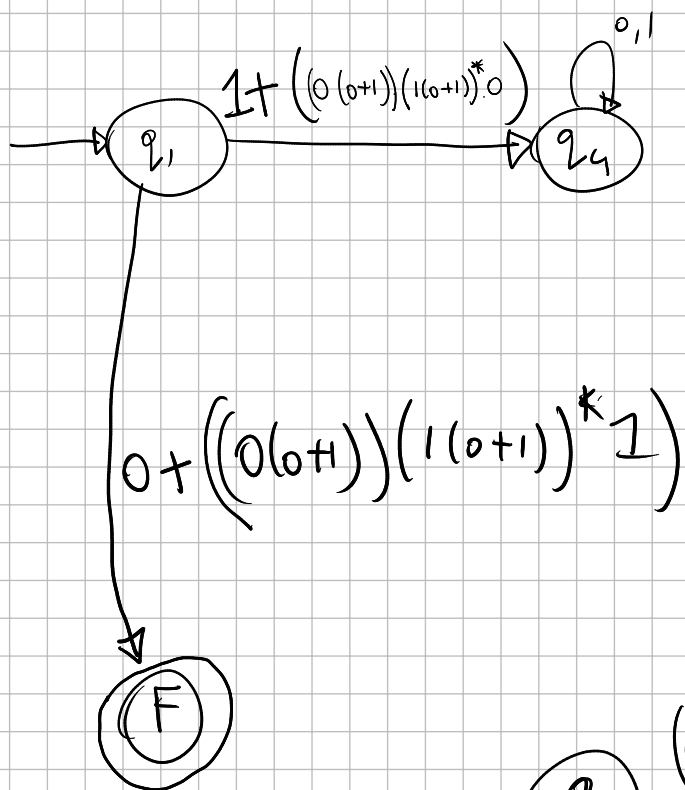
Convert the following DFA into an equivalent regular expression using the state elimination method. First eliminate q_2 , then q_3 and finally q_4 . You must show work.



Eliminate q_2

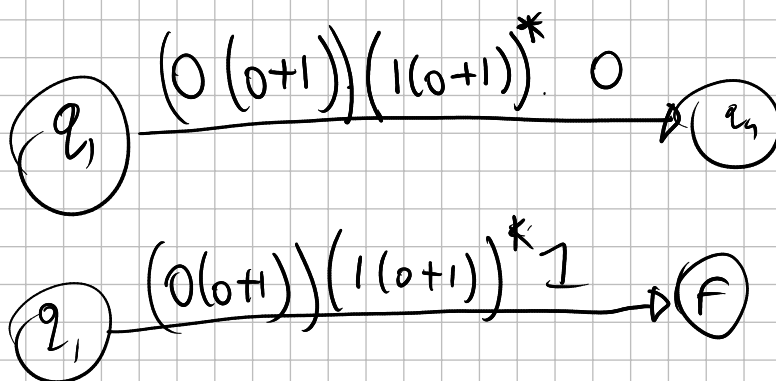
In	out
q_1	q_3
q_3	F

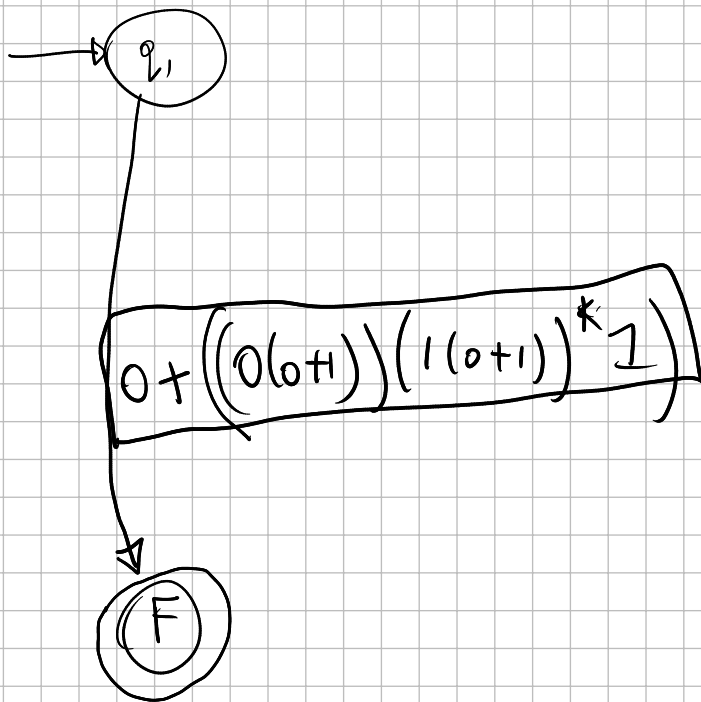




Eliminate q_3

In	out
q_1	q_2
	F





Eliminate q_4

In	out
q_1	X

$$0 + ((0(0+1)) (1(0+1))^* 1)$$